ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Graphs of Functions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

GRAPHS OF FUNCTIONS · CH3 · L5

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The Map

This lesson collapses Graphs of Functions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 34 website questions, 34 checked solutions, 3 local PDFs read.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^2 - \frac{x}{2} - 3$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q15

Solve quadratic equation

$$y = x^2 - \frac{x}{2} - 3$$

The completed table is

x	-3	-2	-1	0	1	2	3
y	7.5	2	-1.5	-3	-2.5	0	4.5

Finish: Missing values are $2, -1.5, -3, 0$.

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 15 (a) Complete the table of values for y x x = - - 2 2 3 x -3 -2 -1 0 1 2 3 y 7.5 -2.5 4.5 (2) (b) On the grid, draw the graph of y x x = - - 2 2 3 for values of x from -3 to 3 -1 -2 O -1 -2 -3 -4 1 2 4 3 2 1 y x 3 -3 5 6 7 8 (2)

YOUR SOLUTION

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x-1)^2 - a$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x-1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y-intercept $(0, 4-a)$, x-intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 20 The curve C has equation y = 4(x - 1)2 - a where a > 4 Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a, (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. y x O

YOUR SOLUTION

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -\frac{2}{x^2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q15

Solve equation

$$y = -\frac{2}{x^2}$$

This is Graph D
This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: D, B, A.

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 15 Here are six graphs. Graph A y x O Graph B y x O Graph C y x O Graph D y x O Graph E y x O Graph F y x O Complete the table below with the letter of the graph that could represent each given equation. Write your answers on the dotted lines. Equation Graph y = 2 2 x y = - 1 2 x3 y = - 5 x

YOUR SOLUTION

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x-1)^2 - a$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x-1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y-intercept $(0, 4-a)$, x-intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 20 The curve C has equation y = 4(x - 1)2 - a where a > 4 Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a, (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. y x O

YOUR SOLUTION

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2x^2 - 4x - 1 = 0$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q21

Read the graph

$$2x^2 - 4x - 1 = 0$$

Read the graph

$$x = -0.2, \quad x = 2.2$$

Finish: (a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 21 Part of the graph of y = 2x2 - 4x - 1 is shown on the grid. -2 O 2 4 -4 -1 1 2 3 y x (a) Use the graph to find estimates for the solutions of the equation 2x2 - 4x - 1 = 0 Give your solutions correct to one decimal place. (2) (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation x2 - x - 1 = 0 Show your working clearly. Give your solutions correct to one decimal place. (3)

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = a\cos bx^{\circ} + c$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q26

The curve is

$$y = a\cos bx^{\circ} + c$$

Use trigonometry

$$3$$

Finish: $\boxed{a = 2,\ b = 3,\ c = 1}$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 26 Here is a sketch of the curve with equation y = a cos bx degrees + c where -90 < x < 450 450 360 270 90 O -90 180 -2 -1 1 2 3 4 y x Find the value of a, the value of b and the value of c a = b = c =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Complete a table of values
"draw", "plot", "construct", a diagram/grid	Plot the curve
"work out", "show", "hence", a familiar exam phrase	Read roots and intersections
"work out", "show", "hence", a familiar exam phrase	Transform the graph
"work out", "show", "hence", a familiar exam phrase	Read the graph
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.